Industrial Internship Report on

"Data Science and Machine Learning Internship"

Prepared by

"Gaurav Kumar"

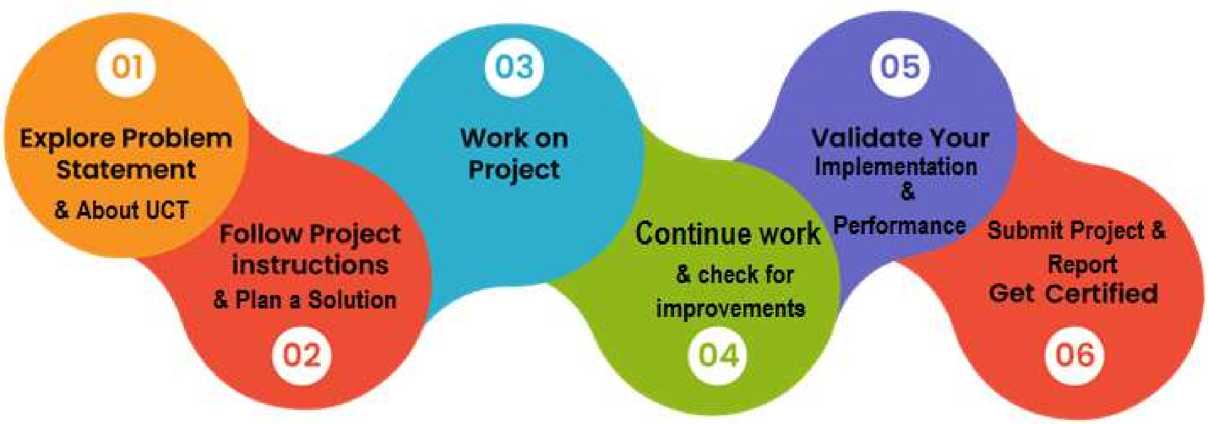
Executive Summary
This report presents the details of the **Data Science and Machine Learning Internship** completed through **UpskillCampus**. The internship was designed to provide practical exposure to real-world industrial problems by applying Data Science and Machine Learning techniques to solve business and engineering challenges. During this internship, I successfully completed two industry-oriented projects. The first project, **Predictive Maintenance of Gearbox Using Vibration Sensors**, focused on analyzing vibration sensor data to predict potential gearbox failures before they occur, enabling timely maintenance and reducing equipment downtime. The second project, **Forecasting Smart City Traffic Patterns**, involved analyzing historical traffic data and building machine learning models to forecast future traffic conditions for efficient traffic management in smart cities. Throughout the internship, I gained practical experience in Python programming, data preprocessing, exploratory data analysis (EDA), feature engineering, machine learning model development, model evaluation, and data visualization. I also learned how to work with real-world datasets, handle missing values, improve model performance, and interpret analytical results. Overall, this internship significantly enhanced my technical knowledge, problem-solving abilities, and understanding of industrial applications of Data Science and Machine Learning. The experience has strengthened my confidence in developing AI-based solutions and has prepared me for future opportunities in the field of Artificial Intelligence, Machine Learning, and Data Science.

TABLE OF CONTENTS

1. Preface:

This report presents my learning journey during the **6-week Data Science and Machine Learning Internship** conducted by **UpskillCampus**. The internship was designed to provide practical exposure to real-world industrial problems and enhance technical knowledge through hands-on implementation of Machine Learning projects.

In today's technology-driven world, practical experience is as important as theoretical knowledge. This internship helped me understand how Data Science and Machine Learning techniques are applied to solve real-world industrial challenges. It also improved my analytical thinking, programming skills, and problem-solving abilities.



During the internship, I successfully completed two industry-based projects:

- **Project 1:** Predictive Maintenance of Gearbox Using Vibration Sensors
- **Project 2:** Forecasting Smart City Traffic Patterns

Both projects involved working with real-world datasets, performing data preprocessing, exploratory data analysis (EDA), feature engineering, machine learning model development, model evaluation, and result interpretation.

The internship was well organized over six weeks. Initially, the problem statements and project objectives were introduced. This was followed by dataset understanding, data cleaning, exploratory analysis, feature engineering, machine learning model development, testing, evaluation, and final project documentation.

Throughout this internship, I gained practical knowledge of **Python, NumPy, Pandas, Matplotlib, Scikit-learn, Machine Learning algorithms, Data Visualization, Feature Engineering, and Model Evaluation**. Apart from technical skills, I also improved my communication, documentation, time management, and project presentation skills.

I sincerely thank **UpskillCampus**, all the mentors, and my college faculty members for their continuous guidance, support, and encouragement throughout this internship. Their valuable suggestions and mentorship helped me successfully complete both projects.

Finally, I would like to encourage my juniors and fellow students to actively participate in such industrial internships. These opportunities provide practical exposure, improve technical skills, and help build confidence for future careers in Artificial Intelligence, Data Science, and Machine Learning.

1 Introduction

1.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

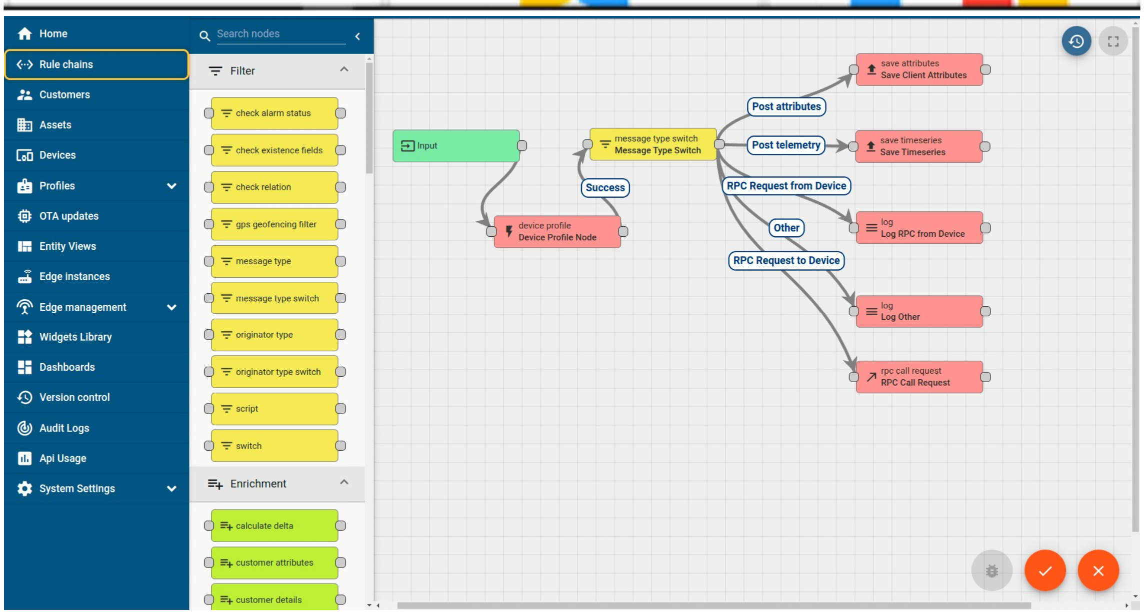
i. UCT IoT Platform (Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

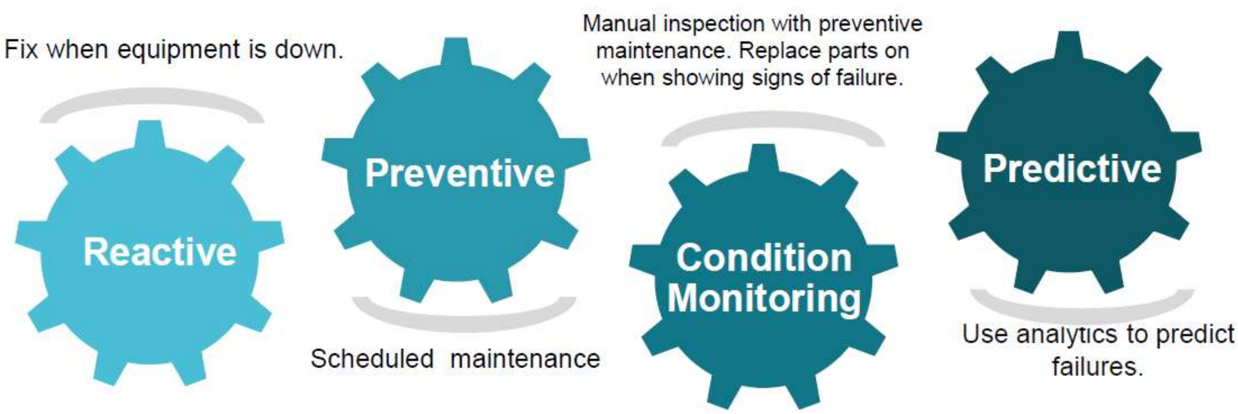
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



1.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

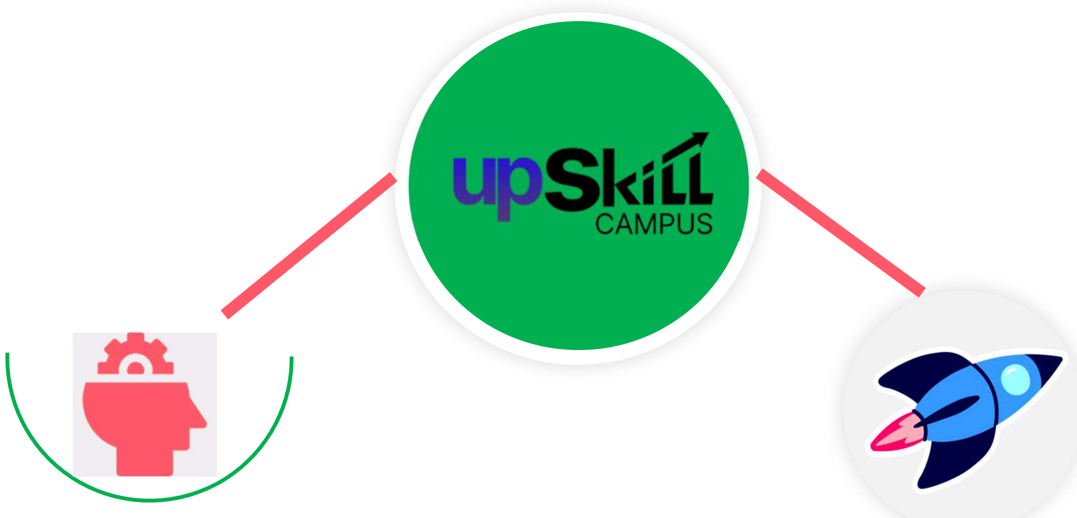
UpskillCampus is an industry-oriented learning platform that focuses on enhancing students' technical and professional skills through internships, certification programs, live projects, and mentorship. The platform collaborates with leading industry partners to provide hands-on learning experiences that bridge the gap between academic knowledge and real-world applications.

The primary objective of UpskillCampus is to prepare students for industry by offering practical exposure in emerging technologies such as Artificial Intelligence, Machine Learning, Data Science, Internet of Things (IoT), Cloud Computing, Cyber Security, and Software Development.

During my internship, UpskillCampus provided a structured learning environment consisting of mentor guidance, project-based learning, regular assessments, and technical support. The internship emphasized solving real-world industrial problems using modern Data Science and Machine Learning techniques.

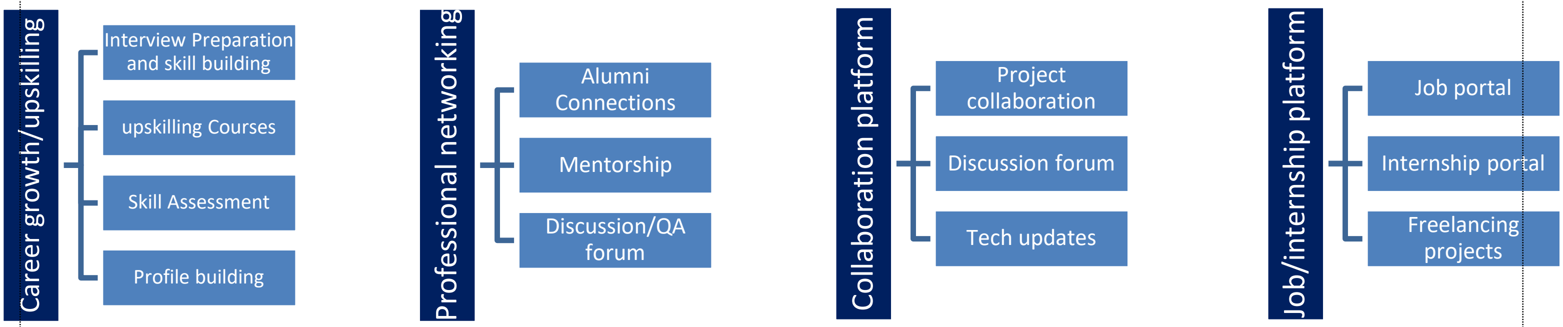
The platform also encouraged continuous learning by providing study resources, project guidelines, and performance evaluations, enabling students to improve their technical skills as well as their communication, teamwork, and problem-solving abilities.

Overall, UpskillCampus played a significant role in helping me gain practical experience, understand industry workflows, and strengthen my foundation in Data Science and Machine Learning.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year



1.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

1.4 Objectives of this Internship program

The primary objectives of this internship were:

- To gain practical exposure to Data Science and Machine Learning concepts.
- To understand the complete Machine Learning project lifecycle.
- To develop proficiency in Python programming for data analysis.
- To perform data preprocessing, exploratory data analysis (EDA), and feature engineering.
- To build, train, and evaluate Machine Learning models using real-world datasets.
- To improve analytical thinking and problem-solving skills.
- To understand industrial applications of predictive analytics.
- To enhance communication, documentation, and project presentation skills.
- To gain experience in working on industry-oriented projects.

1.5 Reference

You can include references such as:

1. Python Documentation – https://docs.python.org/
2. Scikit-learn Documentation – https://scikit-learn.org/
3. Pandas Documentation – https://pandas.pydata.org/
4. NumPy Documentation – https://numpy.org/
5. Matplotlib Documentation – https://matplotlib.org/
6. Kaggle Datasets – https://www.kaggle.com/

1.6 Glossary

Terms	**Meaning**
AI	Artificial Intelligence
ML	Machine Learning
DS	Data Science
EDA	Exploratory Data Analysis
CSV	Comma Separated Values
KPI	Key performance Indicator
IoT	Internet of Things
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
R^2 Score	Coefficient of Determination

3. Problem Statement

3.1 Project 1: Predictive Maintenance of Gearbox Using Vibration Sensors

Problem Statement

Gearboxes are one of the most critical components used in industrial machinery. Continuous operation under heavy loads causes wear and tear, which can eventually lead to unexpected failures. Traditional maintenance approaches, such as reactive maintenance (repair after failure) and preventive maintenance (scheduled maintenance), often result in increased operational costs, unnecessary maintenance activities, and production downtime.

The objective of this project was to develop a **Machine Learning-based Predictive Maintenance System** capable of analyzing vibration sensor data to identify abnormal machine behavior and predict potential gearbox failures before they occur. By using historical vibration data and machine learning algorithms, the system helps industries perform maintenance only when required, thereby improving equipment reliability, reducing maintenance costs, and minimizing unexpected breakdowns.

1.7 3.2 Project 2: Forecasting Smart City Traffic Patterns

1.7.1 Problem Statement

Rapid urbanization has significantly increased the number of vehicles on roads, leading to traffic congestion, longer travel times, higher fuel consumption, and increased environmental pollution. Efficient traffic management has become one of the major challenges faced by modern smart cities.

The objective of this project was to develop a **Machine Learning-based Traffic Forecasting System** capable of predicting future traffic patterns using historical traffic data. The project involved analyzing traffic-related parameters, preprocessing the dataset, performing exploratory data analysis, engineering meaningful features, and training predictive models to forecast traffic volume accurately.

Accurate traffic prediction can assist city authorities in optimizing traffic signal timing, planning road infrastructure, reducing congestion, improving emergency response, and enhancing the overall transportation system.

4. Existing and Proposed solution

4.1 Project 1: Predictive Maintenance of Gearbox Using Vibration Sensors

Existing Solution

In most industries, gearbox maintenance is performed using either **Reactive Maintenance** or **Preventive Maintenance**.

- **Reactive Maintenance:** The gearbox is repaired only after a failure occurs. This often results in unexpected machine downtime, production losses, and high repair costs.
- **Preventive Maintenance:** Maintenance is scheduled at fixed intervals regardless of the actual condition of the gearbox. While this reduces sudden failures, it may lead to unnecessary maintenance and increased operational costs.

Some advanced industries use vibration monitoring systems; however, many traditional systems rely on manual analysis and threshold-based monitoring, which may fail to detect early-stage faults accurately.

Limitations of Existing Solution:

- Unexpected equipment failures in reactive maintenance.
- High maintenance cost due to unnecessary servicing.
- Manual monitoring is time-consuming and prone to human error.
- Threshold-based methods cannot accurately identify complex fault patterns.
- Poor scalability for large industrial environments.

Proposed Solution:

The proposed solution uses **Machine Learning techniques** to analyze vibration sensor data and predict gearbox health. The workflow includes:

- Collecting vibration sensor data.
- Data preprocessing and cleaning.
- Exploratory Data Analysis (EDA).
- Feature Engineering.
- Training Machine Learning models.

- Evaluating model performance.
- Predicting potential gearbox failures before they occur.

This approach enables industries to perform maintenance only when required, improving reliability and reducing downtime.

Value Addition

- Early fault detection.
- Reduced maintenance cost.
- Increased machine reliability.
- Improved production efficiency.
- Data-driven maintenance decisions.
- Better utilization of industrial resources.

4.2 Project 2: Forecasting Smart City Traffic Patterns

Existing Solution:

Most cities rely on **fixed traffic signal timings**, manual traffic monitoring, and historical observations. These approaches cannot effectively adapt to changing traffic conditions caused by weather, holidays, accidents, or peak-hour congestion.

Although some intelligent transportation systems exist, many are expensive to deploy and require specialized infrastructure.

Limitations of Existing Solution:

- Fixed signal timings lead to unnecessary congestion.
- Manual monitoring is inefficient.
- Difficulty handling rapidly changing traffic conditions.
- Limited prediction capability.
- High implementation cost for traditional intelligent traffic systems.

Proposed Solution:

The proposed system applies **Machine Learning algorithms** to historical traffic data to forecast future traffic patterns. The solution includes:

- Data collection.
- Data cleaning.
- Missing value handling.
- Exploratory Data Analysis.
- Feature Engineering.
- Model training and testing.
- Traffic volume prediction.

The generated predictions can support better traffic management and smart city planning.

Value Addition

- Improved traffic forecasting accuracy.
- Better traffic signal management.
- Reduced congestion.
- Lower fuel consumption.
- Reduced environmental pollution.
- Support for smart city decision-making.

4.3 Code submission (Github link)

GitHub Repository:

Project 1: Predictive Maintenance of Gearbox Using Vibration Sensors

Link: https://github.com/gauravkumar902704/UpskillCampus2

Project 2: Forecasting Smart City Traffic Patterns

Link: https://github.com/gauravkumar902704/upskillcampus

4.4 Report submission (Github link) :
https://github.com/gauravkumar902704/UpskillCampus2/blob/main/Reports/final_Report.pdf

5. Proposed Design/ Model

This chapter describes the overall workflow and architecture followed to implement both Machine Learning projects. The solution was developed using a systematic Data Science pipeline consisting of data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and prediction.

5.1 High Level Diagram (if applicable)

Project 1: Predictive Maintenance of Gearbox Using Vibration Sensors

Project 2: Forecasting Smart City Traffic Patterns

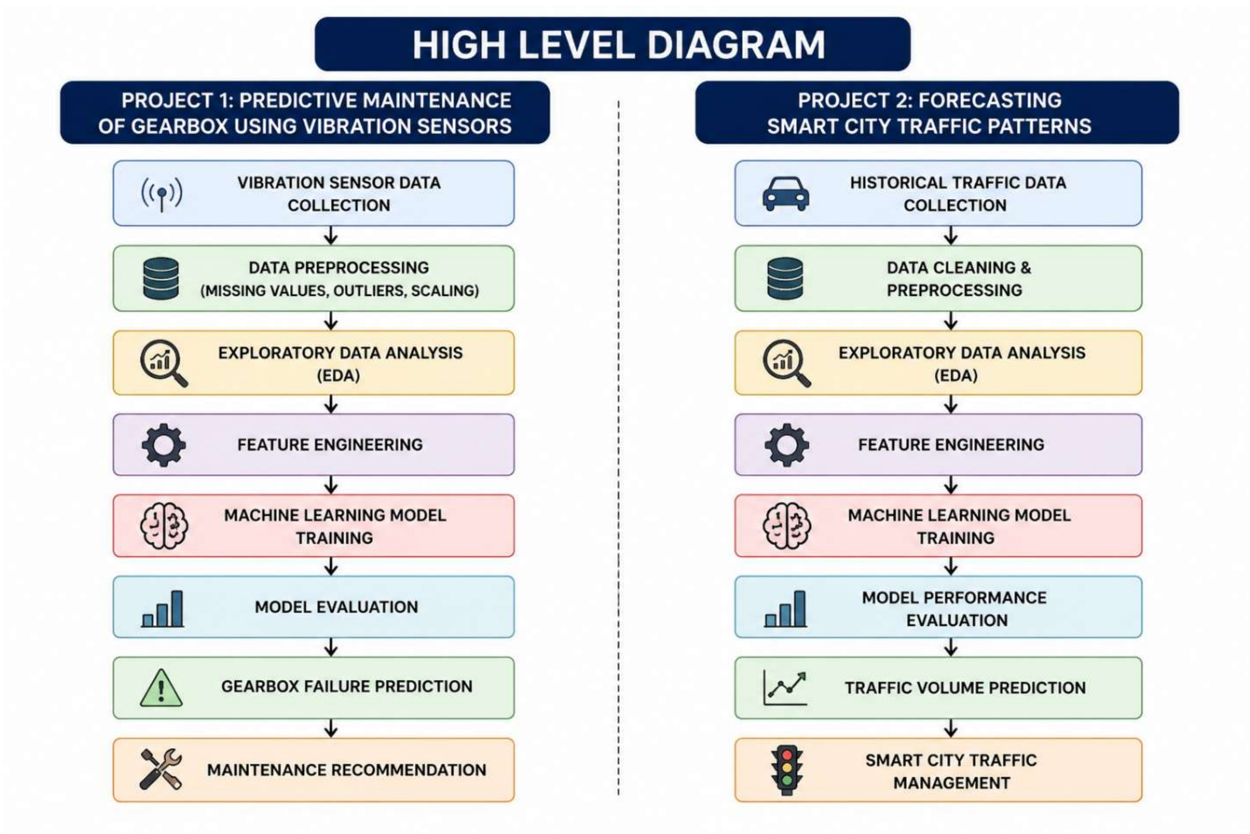


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

Description:

The vibration sensor dataset is provided by UpskillCampus. After cleaning and preprocessing the data, exploratory data analysis is performed to understand feature distributions and relationships. Relevant features are selected before training Machine Learning models. Finally, the trained model predicts whether the gearbox is healthy or likely to fail, enabling predictive maintenance.

5.2 Low Level Diagram (if applicable)

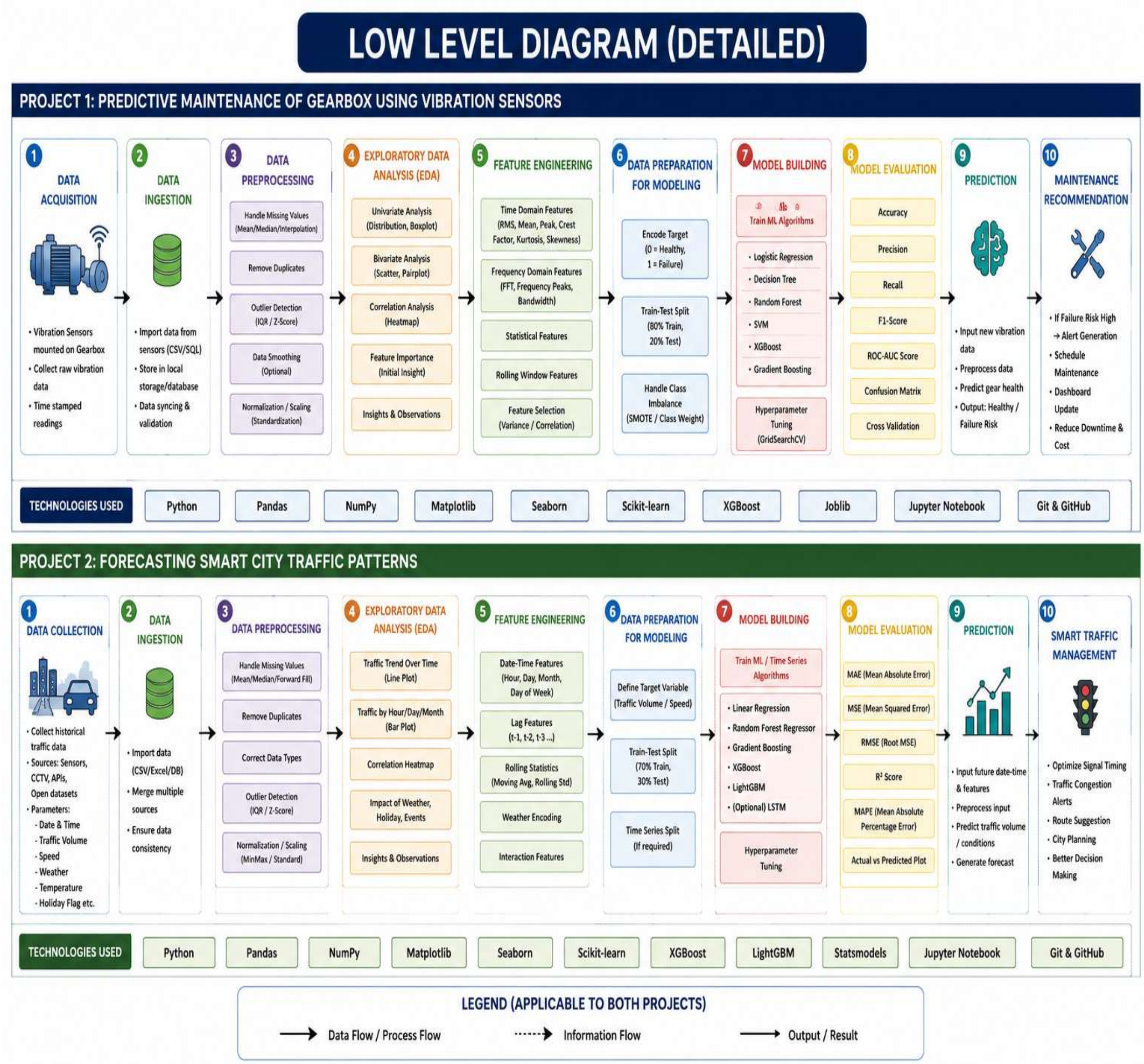


Figure 2 Low Level Daigram

Description

The low-level diagram illustrates the detailed workflow followed during the implementation of both machine learning projects. The process begins with **data acquisition**, where raw data is collected from vibration sensors (Project 1) or historical traffic records (Project 2). The collected data is then imported into the Python environment for further processing.

In the **data preprocessing** stage, missing values, duplicate records, and outliers are handled to improve data quality. The dataset is then normalized or standardized, and categorical variables are encoded wherever required. This step ensures that the data is clean, consistent, and suitable for machine learning algorithms.

After preprocessing, **Exploratory Data Analysis (EDA)** is performed to understand the characteristics of the dataset. Various visualization techniques such as histograms, box plots, scatter plots, correlation heatmaps, and trend analysis are used to identify patterns, relationships, and anomalies in the data.

The next stage is **Feature Engineering**, where meaningful features are created or selected to improve model performance. Statistical, time-domain, frequency-domain, and rolling-window features are extracted for the predictive maintenance project, while date-time, weather, lag, and interaction features are generated for the traffic forecasting project.

The processed dataset is then prepared for model development by splitting it into **training and testing datasets**. Multiple machine learning algorithms are trained, and their hyperparameters are optimized to obtain the best-performing model.

The trained models are evaluated using appropriate **performance metrics**. For the predictive maintenance project, metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix are used. For the traffic forecasting project, MAE, MSE, RMSE, R^2 Score, and MAPE are used to measure prediction accuracy.

Finally, the optimized model is used to generate predictions. In the first project, the system predicts the health condition of the gearbox and recommends maintenance before failure occurs. In the second project, the system forecasts future traffic patterns, enabling efficient traffic management, congestion reduction, and better decision-making for smart city planning.

5.3 Interfaces (if applicable)

Project 1: Predictive Maintenance of Gearbox Using Vibration Sensors

Live Demo Link: **https://gauravkumar902704-project-8-gearbox-predictive-maint-app-lkj6bq.streamlit.app/**

Project 2: Forecasting Smart City Traffic Patterns

Live Demo Link: **gauravkumar902704-smart-city-traffic-forecasting-app-sk4ikf.streamlit.app/**

6. Performance Test:

The performance testing phase was conducted to evaluate the effectiveness, reliability, and prediction capability of the developed Machine Learning models. Both projects were tested using separate training and testing datasets to ensure that the models generalized well to unseen data.

Several evaluation metrics were used to measure the performance of the models. For the Predictive Maintenance project, classification metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix were considered. For the Smart City Traffic Forecasting project, regression metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 Score were used.

Performance testing also involved verifying the correctness of data preprocessing, feature engineering, model training, prediction generation, and evaluation. The results demonstrated that the developed models were capable of making reliable predictions on unseen data while maintaining good performance.

6.1 Test Plan / Test Cases

Project 1: Predictive Maintenance of Gearbox

Test Case ID	Test Scenario	Expected Result	Status
TC-01	Load vibration dataset	Dataset loaded successfully	Passed
TC-02	Handle missing values	Missing values removed/imputed	Passed
TC-03	Remove duplicate records	Duplicate records removed	Passed
TC-04	Perform feature engineering	New useful features created	Passed
TC-05	Train ML model	Model trained successfully	Passed
TC-06	Predict gearbox condition	Correct prediction generated	Passed
TC-07	Evaluate model	Performance metrics displayed	Passed

Project 2: Forecasting Smart City Traffic Patterns

Test Case ID	Test Scenario	Expected Result	Status
TC-01	Import traffic dataset	Dataset imported successfully	Passed
TC-02	Data preprocessing	Clean dataset generated	Passed
TC-03	Exploratory Data Analysis	Visualization generated	Passed

TC-04	Feature Engineering	Features extracted correctly	Passed
TC-05	Train prediction model	Model trained successfully	Passed
TC-06	Forecast traffic volume	Predicted traffic generated	Passed
TC-07	Evaluate model	Error metrics calculated	Passed

6.2 Test Procedure

The following testing procedure was followed for both projects:

1. The dataset was imported into the Python environment.
2. Data quality was verified by checking missing values, duplicate records, and incorrect data types.
3. Data preprocessing techniques such as normalization, encoding, and feature scaling were applied.
4. Exploratory Data Analysis (EDA) was performed to understand the dataset and identify important features.
5. The dataset was divided into training and testing subsets.
6. Machine Learning models were trained using the training dataset.
7. Predictions were generated using the testing dataset.
8. The predicted values were compared with the actual values.
9. Performance metrics were calculated to evaluate model accuracy and prediction capability.
10. The best-performing model was selected for final prediction.

6.3 Performance Outcome

The developed Machine Learning models produced satisfactory results for both projects.

For the **Predictive Maintenance of Gearbox Using Vibration Sensors**, the model successfully identified machine health conditions based on vibration sensor data. The evaluation metrics indicated good classification performance, enabling early detection of potential gearbox failures and reducing unexpected maintenance requirements.

For the **Forecasting Smart City Traffic Patterns** project, the regression model accurately predicted future traffic conditions using historical traffic data. The generated forecasts can assist city authorities in improving traffic signal management, reducing congestion, and supporting data-driven urban planning.

Overall, both projects demonstrated that Machine Learning techniques can effectively solve real-world industrial problems by improving prediction accuracy, reducing manual effort, and enabling informed decision-making.

7 My Learnings

The **Data Science and Machine Learning Internship** at **UpskillCampus** was a valuable learning experience that significantly enhanced my technical knowledge and practical skills. During the internship, I gained hands-on experience in solving real-world problems using data-driven approaches and Machine Learning techniques.

One of the major learnings was understanding the complete **Machine Learning project lifecycle**. I learned how to collect datasets, preprocess raw data, handle missing values, remove duplicate records, detect outliers, and prepare data for analysis. I also gained practical experience in **Exploratory Data Analysis (EDA)**, which helped me understand data distributions, identify patterns, and discover relationships between variables.

The internship strengthened my knowledge of **Feature Engineering**, where I learned how to create meaningful features that improve model performance. I also worked with various Machine Learning algorithms and understood the importance of model selection, training, hyperparameter tuning, and performance evaluation using appropriate metrics.

Through the **Predictive Maintenance of Gearbox Using Vibration Sensors** project, I understood how Machine Learning can be applied in industrial environments to predict equipment failures and reduce maintenance costs. Similarly, the **Forecasting Smart City Traffic Patterns** project helped me understand the practical applications of predictive analytics in smart city infrastructure and transportation systems.

I also improved my programming skills in **Python** and gained practical experience with libraries such as **NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn**. Additionally, I learned to use **Git and GitHub** for project version control and documentation.

Apart from technical knowledge, the internship enhanced my **problem-solving ability, analytical thinking, communication skills, documentation skills, teamwork, and time management**. Working on real-world datasets improved my confidence in handling complex data and developing efficient Machine Learning solutions.

Overall, this internship has strengthened my foundation in **Data Science, Machine Learning, and Artificial Intelligence**. The knowledge and practical experience gained during this internship will help me in my future academic projects, technical interviews, and professional career as an AI/ML Engineer.

8 Future work scope

The projects completed during this internship provide a strong foundation for solving industrial problems using Machine Learning. However, there are several opportunities to further improve and extend these solutions in the future.

For the **Predictive Maintenance of Gearbox Using Vibration Sensors**, future work may include integrating real-time IoT sensor data to enable continuous monitoring of machine health. Advanced Deep Learning models such as LSTM and Transformer-based architectures can be explored for improved fault prediction. The system can also be deployed on cloud platforms with real-time dashboards to support predictive maintenance in large-scale industrial environments.

For the **Forecasting Smart City Traffic Patterns** project, future enhancements may include incorporating live traffic feeds, weather information, GPS data, road accident reports, and special event information to improve prediction accuracy. Time-series forecasting techniques and advanced Deep Learning models such as LSTM, GRU, and Temporal Fusion Transformers can be implemented to capture complex traffic patterns more effectively.

The developed models can also be integrated with smart traffic management systems to optimize traffic signal timing, reduce congestion, improve emergency response, and support intelligent urban planning. Furthermore, deploying these solutions as web applications or cloud-based services would make them accessible to industries, researchers, and government agencies.

Overall, future advancements in Artificial Intelligence, Internet of Things (IoT), and Cloud Computing can further enhance the scalability, accuracy, and real-time applicability of these projects.

9 Thank You Note

I would like to express my sincere gratitude to **UpskillCampus** for providing me with the opportunity to undertake this **Data Science and Machine Learning Internship**. This internship has been an invaluable learning experience that allowed me to apply theoretical concepts to real-world projects and develop practical skills in Data Science and Machine Learning.

I am deeply thankful to all the mentors and trainers at UpskillCampus for their continuous guidance, valuable feedback, and unwavering support throughout the internship. Their expertise, encouragement, and willingness to share knowledge played a significant role in the successful completion of my projects.

I would also like to extend my heartfelt thanks to the **ITM Group of Institutions**, my faculty members, and the Department of Computer Science and Engineering for encouraging me to participate in this internship and supporting my learning journey.

Working on the projects **"Predictive Maintenance of Gearbox Using Vibration Sensors"** and **"Forecasting Smart City Traffic Patterns"** has enhanced my technical knowledge, analytical thinking, and problem-solving abilities. This experience has strengthened my confidence and motivated me to continue learning and building innovative AI and Machine Learning solutions.

Finally, I sincerely thank everyone who directly or indirectly contributed to my internship experience. The knowledge and skills gained during this internship will remain a strong foundation for my future academic pursuits and professional career.

Thank you once again for this wonderful opportunity.

Gaurav Kumar
Data Science and Machine Learning Intern
UpskillCampus